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X-RAY COLLIMATOR WITH IMPROVED TRANSMISSION DEVICE

Abstract

An X-ray collimator has a first rotatable shaft having a first axis arranged longitudinally, a first pair of slats sliding longitudinally and a first transmission device for transmitting motion from the first rotatable shaft to the first pair of slats. The first transmission device has a first belt rotationally jointed with the first rotatable shaft and a first toothed pulley with helical teeth, rotatable about a first secondary axis orthogonal to the first axis and translationally jointed with the slats of the first pair. The first belt is engaged with the first toothed pulley in a semi-crossed manner for transmission of motion between skew axes.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Italian Patent Application No. 102024000005272 filed on Mar. 8, 2024, the contents of which are incorporated by reference in their entirety.

FIELD OF THE INVENTION

[0002] The present invention relates to X-ray equipment, in particular for medical use; even more particularly, the present invention relates to an X-ray collimator with slats for limiting an X-ray beam, provided with an innovative transmission device for moving the slats.

PRIOR ART

[0003] An X-ray collimator is usually applied downstream of a radiogenic source (X-ray tube) and is provided with a first pair of longitudinally translatable slats that relatively approach or move away from one another to narrow or widen the beam passage window longitudinally, and a second pair of transversely translatable slats that relatively approach or move away from one another to narrow or widen the beam passage window transversely.

[0004] To prevent the subject to be X-rayed, for example a patient, from being impacted by X-rays in parts that are not of interest to radiography, it is essential to limit the beam of X-rays as needed; for example, for a knee X-ray, a window of 20×20 centimeters in size is generally sufficient, whilst for a chest X-ray a window of 50×50 centimeters in size is generally required.

[0005] The movement of the two pairs of slats may be performed manually, by rotating two respective knobs, or by means of an electric motor. In both cases, two respective transmission devices allow the transmission of motion from the respective knob or motor to the respective pair of slats.

[0006] To move one of the two pairs of slats, it is necessary to provide two rotatable shafts, arranged orthogonally therebetween. The first shaft is placed rotationally by the knob or by the motor, the second shaft receives the rotary motion from the first shaft by means of a belt and transmits the rotary motion to the slats. For the transmission of motion from the first shaft to the second one, the solutions known nowadays provide for the insertion of a bearing or roller for accompanying the belt onto the pulley of the driven shaft.

[0007] As is understood, the transmission device must be precise and reliable, in such a way that the rotation of a knob corresponds to the approaching and moving away of the slats in a precise and repeatable manner. The Applicant has found however that the transmission of movement between the two orthogonal shafts by means of traditional systems has some drawbacks, especially in the transmission of motion from the belt to the pulley of the driven shaft.

SUMMARY OF THE INVENTION

[0008] The object of the present invention is to implement an X-ray collimator with slats that is provided with a transmission device that meets the needs of the industry and overcomes the drawbacks mentioned hereinabove.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0009] The features and advantages of the X-ray collimator according to the present invention will be apparent from the description hereinbelow, given by way of non-limiting example in accordance with the attached drawings, wherein:

[0010] FIG. 1 shows an X-ray apparatus comprising a collimator according to an exemplary embodiment of the present invention;

[0011] FIG. 2 shows an X-ray collimator according to an exemplary embodiment of the present

invention;

[0012] FIG. 3 shows a mechanism for moving a collimator according to an exemplary embodiment of the present invention, according to a top view;

[0013] FIG. 4 shows the movement mechanism of FIG. 3, according to a bottom view;

[0014] FIG. 5 is a side view of the movement mechanism in FIG. 3; and

[0015] FIG. 6 shows a pulley with helical teeth, according to an exemplary embodiment of the present invention.

DETAILED DESCRIPTION

[0016] With reference to the figures, the numeral 1 has been used to indicate an X-ray apparatus for taking X-rays, mainly in the medical, human or veterinary fields. The apparatus 1 comprises an X-ray machine 2 equipped with an X-ray tube 4 for emitting a beam of X-rays and an X-ray collimator 6 for limiting the width of the beam intended for the user.

[0017] The collimator 6 is applied downstream from the radiogenic source of the X-ray tube, generally by means of an adapter.

[0018] For example, the collimator 6 consists of a box-like body 10 provided with a front face 10a, an upper face 10b, generally provided with a head attachment 12 for the application of the collimator to the X-ray machine, and a lower face 10c provided with an exit window 14 for the collimated beam to exit.

[0019] According to one embodiment, the collimator 6 further comprises a first knob 16 and a second knob 18, which are rotatable in such a way as to modify the dimensions of the exit window 14, preferably arranged upon the front face 10a, in such a way that they are comfortably accessible by the operator.

[0020] According to a further embodiment (not shown), the collimator comprises buttons, generally arranged upon the front face 10a, which may be used by the operator to drive internal motors.

[0021] The collimator 6 further comprises, preferably, a support frame 20 accommodated in the box-like body 10, for supporting the components. The support frame 20 comprises a substantially flat base 22, for example quadrangular in shape, which divides the compartment inside the box-like body 10 into an upper region 24 (visible in FIGS. 3 and 5) and a lower region 26 (visible in FIGS. 4 and 5).

[0022] An opening is defined through the base 22 that is longitudinally divisible by a first pair of slats 32a, 32b, preferably arranged in the lower region 26, and transversely divisible by a second pair of slats 34a, 34b, preferably arranged in the upper region 24. The residual opening determines the exit window 14 of the collimated beam.

[0023] The slats 32a, 32b, 34a, 34b are made of a material with a high X-ray absorption capacity, such as lead.

[0024] The first knob 16 is rotatable about a first axis A1, arranged in the longitudinal direction, and is for example supported by a first shaft 40, preferably arranged in the upper region 24; the rotation in one direction or the other of the first knob 16 respectively determines the moving away from each other or approaching of the slats 32a, 32b of the first pair.

[0025] For this purpose, a first belt 42 is provided that is engaged with the first knob 16 directly or by means of the first shaft 40, for example partially wound thereupon, and a first rotatable secondary shaft 44 is provided that extends along a first secondary axis A1', orthogonal to the first axis A1. The first secondary shaft 44 is preferably arranged in the lower region 26 and is operatively engaged with the first knob 16 by means of the first belt 42, which crosses the base 22.

[0026] In particular, a first toothed pulley 46 with helical teeth is provided that is rotationally jointed with the first secondary shaft 44; the first belt 42 is permanently engaged with the first toothed pulley 46, thus being able to easily transmit the rotary movement of the knob 16 to the first secondary shaft 44. Ultimately, the first belt 42 and the first toothed pulley 46 form a semi-crossed belt drive between skew axes, in particular orthogonal.

[0027] The first belt 42 is preferably toothed and the teeth are trapezoidal and adapted to be

coupled to the helical teeth of the first toothed pulley **46** by shape coupling; during the rotation of the first toothed pulley **46**, the portion of the first belt **42** engaged with the first toothed pulley **46** tends to translate, either forwards or backwards according to the direction of rotation. This ensures particularly fluid motion transmission.

[0028] A further secondary shaft **48** is also provided that is supported in a rotatable manner by the frame **20**, arranged in the lower region **26** and rotatable about a secondary axis $A1''$, parallel to the first secondary axis $A1'$ and arranged opposite to the first secondary shaft **44** with respect to the exit window **14**.

[0029] A first pair of second belts **50a**, **50b** is also provided that connect the first secondary shaft **44** to the further secondary shaft **48**, preferably at the respective transverse ends. The second belts **50a**, **50b** have, respectively, a forward segment (from the first secondary shaft to the further secondary shaft) and a return segment (from the further secondary shaft to the first secondary shaft).

[0030] The first slat **32a** is longitudinally slidable and is attached to the return segments of the two second belts **50a**, **50b**, whilst the second slat **32b** is longitudinally slidable and is attached to the forward segments of the two second belts **50a**, **50b**, in such a way that rotation in one direction of the first knob **16** corresponds to the approaching or moving away from each other of the slats **32a**, **32b** of the first pair and rotation in the other direction of the first knob **16** corresponds to the moving away from each other or approaching of the slats **32a**, **32b** of the first pair.

[0031] Ultimately, the first belt **42**, the first secondary shaft **44**, the first toothed pulley **46**, the further secondary shaft **48** and the first pair of second belts **50a**, **50b** form an example of a first device for transmitting the motion between the first knob **16** and the first pair of slats **32a**, **32b**.

[0032] In one embodiment variant the first transmission device comprises a first electric motor configured to rotationally drive, directly or indirectly, the first shaft **40** in one direction and also in the other direction.

[0033] Furthermore, the second knob **18** is rotatable about a second axis $A2$, arranged in the longitudinal direction, parallel to the first axis $A1$, and is for example supported by a second shaft **50**, preferably arranged in the upper region **24**, on the side opposite to the first shaft **40** with respect to the exit window **14**. The rotation in one direction or the other of the second knob **18** respectively determines the moving way from each other or approaching of the slats **34a**, **34b** of the second pair.

[0034] For this purpose, a second belt **52** is provided that is engaged with the second knob **18** directly or by means of the second shaft **50**, for example partially wound thereupon, and a second rotatable secondary shaft **54** is provided that extends along a second secondary axis $A2'$, parallel to the second axis $A2$. Preferably, the second secondary shaft **54** is arranged in the upper region **24** and is operatively engaged to the second knob **18** by means of the second belt **52**.

[0035] A second toothed pulley **56**, preferably with straight teeth, that is rotationally jointed with the second secondary shaft **54**, for example attached to one end thereof, is permanently engaged with the second belt **52**, thus being able to transmit the rotary movement of the second knob **18** to the second secondary shaft **54**. Ultimately, the second belt **52** and the second toothed pulley **56** form a belt drive between parallel axes.

[0036] A further secondary shaft **58** that is supported in a rotatable manner by the frame **20**, arranged in the upper region **24** and rotatable about a further secondary axis $A2''$, parallel to the second axis $A2$ and to the second secondary axis $A2'$, and arranged opposite to the second secondary shaft **54** with respect to the exit window **14**, is also engaged with a second pair of second belts **60a**, **60b**. The second belts connect the second secondary shaft **54** to the further secondary shaft **58**, preferably at the respective longitudinal ends thereof, and have a forward segment (from the second secondary shaft to the further secondary shaft) and a return segment (from the further secondary shaft to the second secondary shaft), respectively.

[0037] The first slat **34a** of the second pair is transversely slidable and is attached to the return segments of the two belts **60a**, **60b** of the second pair, whilst the second slat **34b** is transversely

slidable and is attached to the forward segments of the two belts **60a**, **60b** of the second pair, whereby the rotation in one direction of the second knob corresponds to the approaching or moving away from each other of the slats **34a**, **34b** of the second pair and the rotation in the other direction of the second knob corresponds to the moving away from each other or approaching of the slats **34a**, **34b** of the second pair.

[0038] Ultimately, the second belt **52**, the second secondary shaft **54**, the second toothed pulley **56**, the further secondary shaft **58** and the second pair of second belts **60a**, **60b** form an example of a second device for transmitting the motion between the second knob **18** and the slats **34a**, **34b** of the second pair.

[0039] In an embodiment variant the second transmission device comprises a second electric motor configured to rotationally drive, directly or indirectly, the second shaft **50** in one direction and also in the other direction.

[0040] Innovatively, the collimator according to the present invention meets the needs of the sector and overcomes the aforementioned drawbacks insofar as the elasticity of the first belt, in combination with the helical teeth of the first toothed pulley, allows the motion between orthogonal shafts to be transmitted precisely and repetitively.

[0041] Advantageously, moreover, the teeth of the first belt and the consequent translation of the engaged portion, allow motion between the first knob and the first secondary shaft to be transmitted in a very fluid manner.

[0042] It is understood that a person skilled in the art, in order to meet contingent needs, may make modifications to the device described above, wherein all thereof are contained within the scope of protection as described and claimed herein.

Claims

1. An X-ray collimator for delimiting an exit window of an X-ray beam, comprising a first rotatable shaft having a first axis arranged longitudinally, a first pair of slats sliding longitudinally, approaching or moving away from each other to longitudinally delimit the exit window, and a first transmission device for transmitting motion from the first rotatable shaft to the first pair of slats, wherein the first transmission device comprises a first belt rotationally jointed with the first rotatable shaft, a first toothed pulley with helical teeth, rotatable about a first secondary axis orthogonal to the first axis and a first transmission mechanism connected to the first toothed pulley and configured to make the slats of the first pair of slats move closer together or further apart according to a rotation of the first toothed pulley, the first belt being engaged with the first toothed pulley in a semi-crossed manner for transmission of motion between skew axes.
2. The X-ray collimator of claim 1, wherein the first belt is toothed.
3. The X-ray collimator of claim 2, wherein the first belt has a trapezoidal toothing adapted to be coupled to the first toothed pulley by shape coupling.
4. The X-ray collimator of claim 1, wherein a segment of the first belt, gripped by the first toothed pulley, is translatable on the first toothed pulley during the rotation of the first toothed pulley.
5. The X-ray collimator of claim 1, wherein the first transmission device further comprises: a first secondary shaft rotatable about the first secondary axis, the first toothed pulley being rotationally jointed with the first secondary shaft; a further secondary shaft rotatable about a secondary axis parallel to the first secondary axis, the further secondary shaft being arranged opposite to the first secondary shaft with respect to the exit window; and a first pair of second belts for transmitting motion between the first secondary shaft and the further secondary shaft, arranged on transversely opposite sides of the exit window, wherein one slat of the first pair of slats is fixed to a respective forward segment of the second belts of the first pair of second belts and the other slat of the first pair of slats is fixed to a respective return segment of the second belts of the first pair of second belts.

- 6.** The X-ray collimator of claim 1, further comprising a second rotatable shaft having a second axis arranged longitudinally parallel to the first axis, a second pair of slats sliding transversely, approaching or moving away from each other to transversely delimit the exit window, and a second transmission device for transmitting motion from the second rotatable shaft to the second pair of slats, wherein the second transmission device comprises a second belt rotationally jointed with the second rotatable shaft, a second toothed pulley, rotatable about a second secondary axis parallel to the second axis, and a second transmission mechanism connected to the second toothed pulley and configured to make the slats of the second pair of slats move closer together or further apart according to a rotation of the second toothed pulley, the second belt being engaged with the second toothed pulley for transmitting motion between parallel axes.
- 7.** The X-ray collimator of claim 6, wherein the second toothed pulley has straight teeth.
- 8.** The X-ray collimator of claim 6, wherein the second transmission device further comprises: a second secondary shaft rotatable about the second secondary axis, the second toothed pulley being rotationally jointed with the second secondary shaft; a further secondary shaft rotatable about a further secondary axis parallel to the second secondary axis, the further secondary shaft being arranged opposite to the second secondary shaft with respect to the exit window; and a second pair of second belts for transmitting motion between the second secondary shaft and the further secondary shaft, arranged on longitudinally opposite sides of the exit window, wherein one slat of the second pair of slats is fixed to a respective forward segment of the second belts of the second pair of second belts and the other slat of the second pair of slats is fixed to a respective return segment of the second belts of the second pair of second belts.
- 9.** The X-ray collimator of claim 8, further comprising a support frame having a substantially flat base identifying an upper region and a lower region.
- 10.** The X-ray collimator of claim 9, wherein the first shaft is arranged in the upper region, the first secondary shaft with the first toothed pulley in the lower region), and the first belt crosses the flat base.
- 11.** The X-ray collimator of claim 10, wherein the further secondary shaft of the first transmission device and the first pair of second belts are arranged in the lower region.
- 12.** The X-ray collimator of claim 9, wherein the second shaft and the second secondary shaft with the second toothed pulley are arranged in the upper region.
- 13.** The X-ray collimator of claim 12, wherein the further secondary shaft of the second transmission device and the second pair of second belts are arranged in the upper region.
- 14.** The X-ray collimator of claim 1, wherein the X-ray collimator is manually operable, optionally by knobs.
- 15.** The X-ray collimator of claim 1, wherein the X-ray collimator is electrically operable, optionally by electric motors.
- 16.** An X-ray apparatus comprising: an X-ray machine provided with a ray tube for emitting an X-ray beam; and the X-ray collimator of claim 1, applied to the ray tube, downstream of an X-ray source.
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